Título do trabalho

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Contents

Chapter 1

Introduction

High-performance computing (HPC) has become essential to modern science, and its capability has grown by drawing more power, packing more hardware, and cycling through that hardware ever faster (Portegies Zwart, 2020; Allen, 2022). The environmental cost of this growth has traditionally been equated with operational energy consumption (Koomey *et al.*, 2011), yet operational energy is only one facet of the footprint. Manufacturing the processors, memory, and interconnects that populate a machine emits carbon long before the system is switched on, and this embodied share is rising as fabrication grows more complex (Gupta *et al.*, 2021; Pirson *et al.*, 2023). Cooling those systems and generating the electricity that feeds them consumes freshwater at scales large enough to matter regionally (Mytton, 2021). The semiconductor supply chain draws on scarce materials and discharges hazardous waste (Sandhu *et al.*, 2025), while the shortening replacement cycles that keep systems competitive turn still-functional hardware into electronic waste. Across its full lifecycle, then, an HPC system carries costs in carbon, water, and materials and leaves electronic waste behind, well beyond the operational energy it draws.

Visibility into this broader footprint has only recently begun to grow, and the established ways of evaluating and operating HPC systems still capture only part of it. The Green500 ranks systems by performance per watt under a single benchmark (Feng and Cameron, 2007), reducing sustainability to an efficiency ratio measured on Linpack. Job schedulers decide when and where computation runs, yet they have conventionally been designed to prioritize performance above all else (Kocot *et al.*, 2023; Czarnul *et al.*, 2019), leaving environmental cost outside the optimization objective. Procurement and hardware-replacement decisions weigh performance and acquisition cost while treating lifecycle impact as an afterthought (Li *et al.*, 2023). Each of these approaches sharpens a single dimension of a problem whose full extent remains only partially charted, and the gains they deliver are repeatedly absorbed by growing demand, so the best-measured dimension does not improve in absolute terms (York *et al.*, 2022). Attention to the dimensions these approaches leave out is a comparatively recent and still-emerging area of interest.

This work begins with a systematic mapping study that turns this emerging interest into a concrete gap (Rosa *et al.*, 2026). Classifying 62 reports from sixteen years of research across environmental impacts, lifecycle stages, system loci, and management levers, the study finds the literature concentrated in a single corridor of that classification: carbon among the impacts, operations among the lifecycle stages, facility infrastructure among the system loci, and measurement among the management levers. The dimensions outside that corridor sit in sparsely populated cells, studied rarely and in isolation: water beyond cooling, materials beyond what carbon accounting already captures, biodiversity, and end-of-life processing. The rest of this work sets out to fill the gaps the map identifies, and it currently runs along two branches.

The first branch focuses on HPC systems' operation. The batch scheduler decides when and where jobs run on performance grounds, users submit work without seeing its environmental cost, and hardware-replacement decisions weigh performance and price while overlooking embodied and end-of-life impacts. This work responds by building simulation infrastructure to test a range of mechanisms: scheduling heuristics that react to a time-varying grid, user-facing accounting that makes cost visible, platform controls that trade performance for a smaller footprint, and lifecycle-aware policies that tie hardware aging and replacement to an environmental budget. The infrastructure and its footprint model track operational carbon, water, and embodied impact, so this work can study these mechanisms without access to a production system. A first result along this line is Greenfilling, a small change to EASY backfilling (Skovira *et al.*, 1996) that holds back opportunistic job starts when the targeted grid intensity, carbon or water, runs above its recent average. Evaluated across three national grids and two production machines, it bought certain delay for uncertain environmental return, which sets a lower bound on how sophisticated such an intervention has to be. Its successor, job shifting, prices delaying each job against the grid intensities it would meet, and a static analysis of the same workloads and signals places its ceiling near 7% of operational carbon, two orders of magnitude above what Greenfilling reached.

The second branch focuses on HPC systems' impacts as a whole. The Green500 ranks systems by performance per watt, but this is just one side of the real impacts of using those platforms. Two systems can rank side by side while drawing very different absolute power, sitting on grids whose water and carbon intensity differ by orders of magnitude, carrying different embodied impacts, and consuming water at rates that do not match the regional scarcity. The aim is not to replace the Green500 but to use it as a reference point, showing what a sufficiency-aware lens reveals: cases where efficiency improves while absolute burden grows, where embodied costs dominate but stay invisible, and where regional context changes which system is actually less harmful. A first result along this line is an initial literature analysis that identified four candidate layers for such a lens and mapped the evidence coverage across them.

The remainder of this document is organized as follows. Chapter 2 reviews the environmental footprint of HPC across its lifecycle. Chapter 3 presents the systematic mapping study and the classification behind the gap this work addresses. Chapter 4 reports the preliminary results on both branches. Chapter 5 sets out what remains. The first branch adds user modeling, platform controls, and lifecycle-aware policies to the scheduling mechanism tested so far. The second develops a sufficiency-aware framework and builds an empirical case, from modeled and real systems, for asking whether and how much additional computation is environmentally justified. Chapter 6 closes with the schedule and the expected contributions of the completed work.

Chapter 2

The Environmental Footprint of HPC

Chapter 3

What the Field Measures: A Systematic Mapping

Chapter 4

Preliminary Results

4.1 Accounting for Environmental Cost in HPC Operations

4.1.1 Greenfilling: Suppressing Backfill Starts on a Relative Signal

The batch scheduler is a practical place to act on operational impacts because it already decides when and where every job runs, but an environmental gain from scheduling is only worth having if the performance penalty stays acceptable. EASY backfilling is a well-established baseline for weighing that trade-off. It keeps a capacity reservation for the job at the head of the queue and raises utilization by filling the resulting gaps with later jobs that do not delay that reservation (Skovira *et al.*, 1996; Mu'alem and Feitelson, 2001). The underlying question is whether a simple scheduling heuristic can lower an HPC center's carbon or water footprint without hurting performance. Greenfilling is a first attempt at answering it, and the smallest intervention we could make against the EASY baseline: it leaves the reservation untouched and modifies exactly one decision, permitting a backfill start only when the targeted grid intensity, carbon or water, observed at the scheduling instant does not exceed its exponential moving average. Building and evaluating it doubled as a first exercise of the simulation infrastructure this branch needs.

The rule is event-driven: Greenfilling is implemented as a Batsched variant that subclasses the EASY implementation it is compared against, and the scheduler runs at every job arrival and completion. At the k-th such event, at time t_k, it reads the targeted intensity $I(t_k)$ from the 15-minute trace and updates a smoothed reference

$$S_k = \alpha\, I(t_k) + (1 - \alpha)\, S_{k-1}, \qquad S_0 = I(t_0), \tag{4.1}$$

with smoothing factor $\alpha = 0.5$ throughout the campaign. The scheduler permits backfill starts at t_k if and only if $I(t_k) \le S_k$, and it schedules the reservation for the job at the head of the queue regardless of this condition. Because Equation (4.1) advances once per scheduling event rather than once per trace step, the effective memory of S_k tracks the density of arrivals and completions instead of wall-clock time. When many events fall inside one 15-minute step the reference converges toward the current sample, so how long backfill stays suppressed after an upward step of the signal depends on how busy the machine is, not only on α.

Evaluating Greenfilling first required simulation support that existing toolchains do not offer, since current simulators do not track time-varying carbon and water footprints and a recent extension in this direction covers only operational carbon emissions (Saraiva *et al.*, 2025). We therefore extended SimGrid (Casanova *et al.*, 2025) and Batsim (Dutot *et al.*, 2017) with a per-host footprint model that accumulates operational carbon from the grid carbon intensity, on-site water through the water usage effectiveness (WUE), off-site water through the power usage effectiveness (PUE) and the grid water intensity, and embodied carbon and water as separate values. All four operational parameters can vary independently over time, Batsim replays timestamped updates to them during scheduling experiments, and each job's operational and embodied footprints are exported separately. The user-facing accounting, platform controls, and lifecycle-aware policies still to come will build on this same model.

The grid signals derive from ENTSO-E actual net generation by production type (Hirth *et al.*, 2018), resampled to a 15-minute grid and converted into generation-weighted carbon and water intensities using lifecycle carbon factors from UNECE where available and IPCC otherwise (United Nations, 2022; Intergovernmental Panel On Climate Change, 2015) and operational water consumption factors (Macknick *et al.*, 2012). Germany, France, and Poland provide three contrasting signals from the same source and observation period, dominated by hard coal and lignite in Poland, by nuclear in France, and by a more mixed and changing supply in Germany. The contrast is not one-dimensional: France has the lowest mean carbon intensity of the three but the highest mean water intensity, 36.4 gCO_2eq/kWh and 3.320 L/kWh against Poland's 569.4 gCO_2eq/kWh and 1.587 L/kWh. A consistency check against Ember's independent production-based national intensities for 2018 to 2024 yields Pearson correlations between annual means of 0.987 for Germany, 0.994 for France, and 0.999 for Poland, with modeled levels 16.9%, 34.0%, and 19.7% lower, a gap that the choice of emission factors largely explains. Since Greenfilling reacts to changes over time rather than to absolute levels, the preserved ordering and annual trend matter more here than the level offsets.

The evaluation campaign pairs these signals with two LANL machines of different scales, Mustang at 1,600 nodes and Trinity at 9,408 nodes, each represented by one slack and one stress workload extract of four weeks selected on utilization and queue conditions. Each country contributes 12 environmental windows of 28 days, three drawn per season under a fixed seed, for 36 windows in total. Crossing windows, machines, workload regimes, and three schedulers, EASY plus a carbon-targeting and a water-targeting Greenfilling, gives 432 simulations, and pairing each Greenfilling run with the EASY run of the same window, machine, and regime gives 288 paired comparisons. Table 4.1 summarizes the design. The platform model is deliberately simple: jobs are rigid and replayed at their recorded size, nodes draw 10 W idle and a constant 320 W or 270 W while busy on Mustang and Trinity, and the facility defaults to a PUE of 1, a WUE of 0, and no embodied values, so the campaign reports operational grid carbon and off-site water alone. Constant active power also bounds what scheduling can reach, since reordering the same jobs cannot change active energy and the only remaining levers are idle periods and the operational intensities a job encounters while it runs. We call Greenfilling competitive when it reduces the targeted operational footprint relative to EASY without increasing mean bounded slowdown.

Varied factors	
Grid signal country	Germany, France, Poland
Environmental window	12 per country, 3 per season, 28 days
Machine	Mustang (1,600 nodes), Trinity (9,408 nodes)
Workload regime	slack, stress
Scheduler	EASY, carbon Greenfilling, water Greenfilling
Fixed settings	
EMA smoothing factor	$\alpha = 0.5$
Grid signal resolution	15 minutes
Node power	10 W idle, 320/270 W busy (Mustang/Trinity)
Facility model	PUE = 1, WUE = 0, no embodied impacts
Total	432 runs, 288 paired comparisons

Table 4.1: *Parameters of the Greenfilling evaluation campaign.*

Greenfilling is not competitive in any evaluated scenario. Figure 4.1 shows the paired changes relative to EASY, and Table 4.2 breaks the medians down by machine, regime, and objective. No scenario median footprint change reaches 0.1% in magnitude, for carbon or for water, while mean bounded slowdown rises in every one of the 288 pairs, between +24.5% and +757.2%, with scenario medians between +161.5% and +323.6%. The 95th-percentile waiting time of replayed jobs also rises in all 288 pairs. Even the most favorable case, Mustang under the slack regime, where carbon falls in 30 of 36 windows under carbon Greenfilling and water falls in 29 of 36 under water Greenfilling, buys those reductions at a median slowdown increase of +323.6%. Under stress the Mustang medians turn slightly positive and Trinity stays near zero in both regimes. Zero of the 288 pairs reduce the targeted footprint without increasing mean bounded slowdown.

The energy column of Table 4.2 separates the two ways a footprint can change under this platform model, consuming a different amount of energy or meeting a different intensity while consuming it. On Mustang under slack and on Trinity in both regimes the consumed energy does not move, so the footprint change there is purely an exposure change, and that exposure change stays below 0.1%. Mustang under stress is the one exception. Median platform energy rises by 0.043% under carbon Greenfilling and 0.044% under water Greenfilling, while the effective targeted intensity, the targeted operational footprint divided by consumed energy, falls by 0.037% and 0.032%. The rule does buy a small exposure improvement there, but the extra energy from stretching the schedule cancels it, which is how the stress medians on Mustang end up slightly positive. The campaign accounts energy at the whole-platform level, so this decomposition shows the mechanism but cannot attribute the extra energy to individual jobs.

The design itself points at why deferral did not translate into lower footprints. Greenfilling decides whether a backfill may start at the current instant, but it never selects a later

Machine	Regime	Objective	Footprint	Energy	Slowdown
Mustang	slack	carbon	−0.068%	0.000%	+323.6%
		water	−0.031%	0.000%	+308.5%
	stress	carbon	+0.008%	+0.043%	+175.0%
		water	+0.014%	+0.044%	+161.5%
Trinity	slack	carbon	−0.005%	0.000%	+210.1%
		water	−0.007%	0.000%	+211.5%
	stress	carbon	+0.001%	0.000%	+226.6%
		water	−0.000%	0.000%	+213.3%

Table 4.2: *Median paired changes relative to EASY by machine, workload regime, and Greenfilling objective. The footprint column reports the signal each objective targets. On Mustang under slack the consumed platform energy is identical to EASY in every window, and on Trinity it is unchanged to numerical precision at the median.*

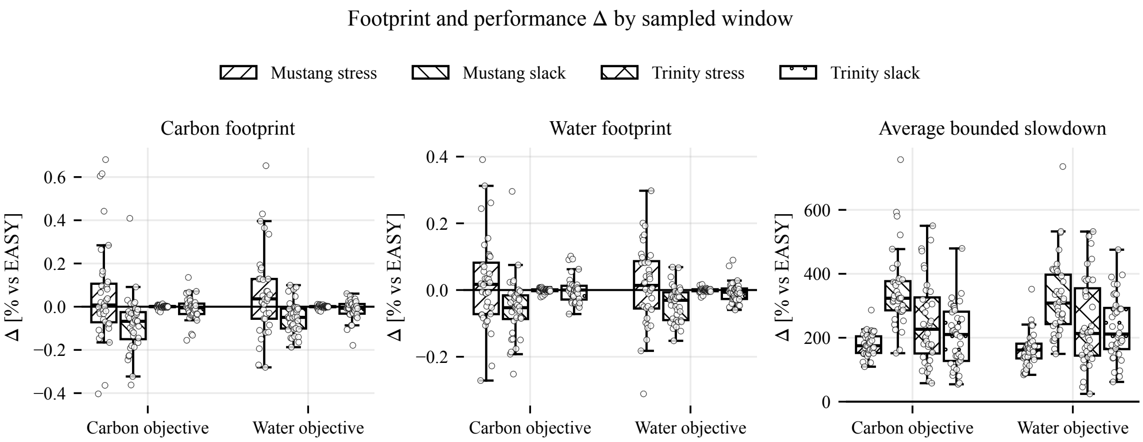

Figure 4.1: *Footprint and performance change relative to EASY across the 288 paired comparisons. Carbon and water footprint changes stay small, with medians below 0.1% in magnitude, while mean bounded slowdown worsens in every pair.*

start based on predicted intensity. Under constant active power, postponing a job changes its footprint only if its execution actually moves to lower-intensity intervals, and the campaign does not determine whether that condition failed because of the decision rule, the temporal structure of the signals, the workloads, or their interaction. The campaign does probe the signal hypothesis once. Each window's swing, the mean daily peak-to-trough range of the targeted signal divided by the window mean, correlates only weakly with the achieved saving, at a pooled Pearson $r = 0.079$ for carbon and $r = 0.247$ for water over 144 pairs per objective, with per-scenario coefficients between −0.09 and 0.59 and no consistent sign. Every window is reused by all four machine and regime scenarios, so the pooled points are not independent, but windows with more intra-day variation did not systematically yield larger savings. The result is therefore narrow: under the evaluated signals, workloads, and platform assumptions, suppressing EASY's backfill starts on a moving-average comparison does not reduce the targeted footprint enough to offset its increase in mean bounded slowdown.

Two lessons from the campaign now shape the mechanism that replaces it. The first is the asymmetry between refusing a start and choosing one. A suppressed backfill start buys a certain queueing delay for an environmental return that depends on the intensities the deferred job eventually meets, and the campaign shows how one-sided that bet is. The second is that a relative threshold and a relative metric compound each other. The rule fires on the same relative deviation in France as in Poland even though their mean carbon intensities differ by more than an order of magnitude, a 0.1% saving reads identically in both, and the quantity that would distinguish them, the absolute footprint avoided per unit of scheduling penalty, was not measured. Together, these lessons set a lower bound: an intervention that only refuses starts on a relative signal is too little. A viable heuristic has to price both sides of each decision, the footprint of starting a job now against the footprint of waiting, and it has to weigh absolute rather than relative stakes.

The campaign also produced reusable infrastructure. We release the intensity traces, workload extracts, campaign configuration, and analysis as a reproducible artifact,[1] and we submitted this analysis to the EESP 2026 workshop.

4.1.2 Job Shifting: Pricing Each Start in Absolute Terms

The Greenfilling lessons point to job shifting as the next mechanism to explore. Rather than suppressing opportunistic starts on a relative comparison, the scheduler could choose each job's start time by pricing every candidate start within a bounded delay, in the absolute units of the targeted impact.

The model prices a single job: let j be a job with execution time p_j, q_j allocated nodes, and submission time r_j, and let I be the time-varying intensity of the targeted signal. Starting j at a time $t \geq r_j$ costs

$$C_j(t) = P_{\text{idle}}\, q_j \int_{r_j}^{t} I(\tau)\,\mathrm{d}\tau + P_{\text{comp}}\, q_j \int_{t}^{t+p_j} I(\tau)\,\mathrm{d}\tau, \tag{4.2}$$

where P_{idle} and P_{comp} are the platform's idle and busy node powers. The first term charges for the wait: while j holds back, its q_j nodes sit idle and accumulate impact at the intensities of the interval $[r_j, t)$. The second term charges for the run, integrating the intensity over the window the execution would actually occupy rather than sampling it at the start. The heuristic selects the start t^* that minimizes C_j over $r_j \leq t \leq r_j + d_m$, where the maximum displacement d_m bounds how long any job can be held. At $t = r_j$ the waiting term vanishes and Equation (4.2) reduces to the cost of starting immediately, so not shifting is itself a candidate and the heuristic only moves a job when the trace ahead pays for the wait. Carbon and water each define their own cost function through their own intensity series, and thus their own t^*. The cost of deferral appears as idle exposure, the return appears as the difference in integrated intensity over the execution window, and both sides are absolute quantities rather than deviations from a moving average.

Before the heuristic reaches the scheduler, the same cost function supports a static analysis of the workloads and traces. The grid signals are step functions on a 15-minute

[1] https://github.com/Lucas-Doctorate-Project/experimental-artifacts/tree/paper

grid, so the integrals in Equation (4.2) collapse into sums. We use this to compute a ceiling: for each job, excluding the warm-up jobs that reconstruct the machine state at the start of each extract, we record the saving it would collect if it could shift freely within d_m, and we accumulate these per-job gains over the same 144 workload and window combinations as the campaign, with the same platform power values, for d_m between zero and 24 hours. The ceiling is optimistic by construction, since summing independent per-job gains ignores node capacity and queue contention and lets every job claim the same low-intensity valleys at once. Its value is diagnostic: if even this bound were small, no shifting heuristic on these signals would be worth implementing.

The bound is not small, as Figure 4.2 shows. The aggregate ceiling grows with the allowed displacement and reaches 6.76% for carbon and 4.89% for water at a d_m of 24 hours, almost two orders of magnitude above the sub-0.1% medians of the Greenfilling campaign. Water pays off less than carbon over the long run, likely due to its lower day-to-day variability. Averaged over the 12 windows per country used in the Greenfilling campaign, mean swing is 0.50 for carbon against 0.47 for water in Germany, 0.32 against 0.29 in France, and 0.24 against 0.23 in Poland (Table 4.3).

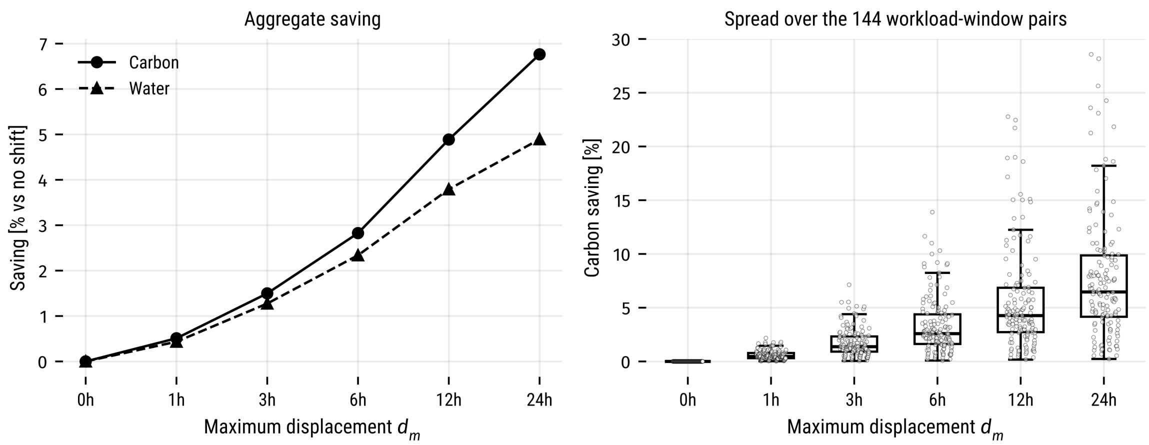

Figure 4.2: *Ceiling of the static job-shifting analysis as a function of the maximum displacement d_m. The left panel shows the aggregate saving over all workloads and windows for each signal. The right panel shows the spread of the carbon saving across the 144 workload and window pairs.*

Grid	Signal	Mean	Median	Std	Min	Max
Germany	carbon	0.50	0.45	0.22	0.29	0.93
	water	0.47	0.39	0.21	0.27	0.84
France	carbon	0.32	0.30	0.08	0.18	0.51
	water	0.29	0.27	0.07	0.19	0.44
Poland	carbon	0.24	0.20	0.15	0.09	0.61
	water	0.23	0.19	0.15	0.10	0.60

Table 4.3: *Swing of the carbon and water signals by grid, over the 12 windows per country used in the Greenfilling campaign.*

Within that bound, the gains concentrate by grid and by season (Figure 4.3). The

German signal offers the most, with an aggregate ceiling of 11.69% at 24 hours, against far less in France and Poland, and the seasons follow the same mechanism: spring reaches 12.32% against winter's 4.89%. The gap tracks the same swing metric: over the three windows drawn from each of the three grids, mean carbon swing is 0.50 in spring against 0.28 in winter, nearly double, so a job displaced in spring meets a wider daily range between low and high intensity than one displaced in winter, and that wider range is what the heuristic exploits. The four workload scenarios behave similarly, which points at the signal rather than the workload as the driver, though with only three windows per grid and season the seasonal split is still a tendency rather than a settled result.

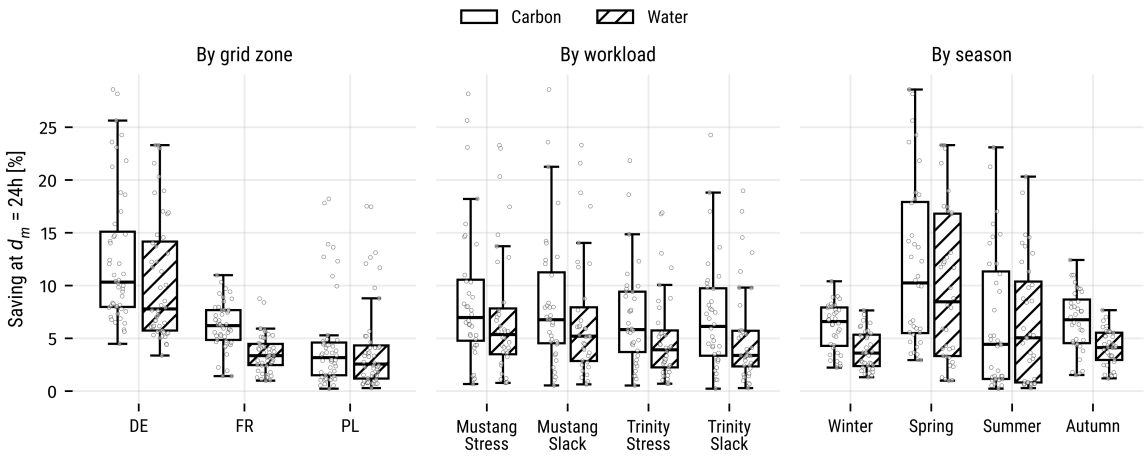

Figure 4.3: *Ceiling at a d_m of 24 hours across the 144 workload and window pairs, split by grid zone, workload scenario, and season.*

Only a fraction of jobs actually move (Figure 4.4): at a d_m of 24 hours, 22.2% of jobs are displaced at all, with a median displacement of 9.25 hours among those that move, while the other 78% already start at the best point they can reach. The aggregate ceiling is therefore carried by roughly a fifth of the jobs, which sharpens the capacity caveat, since the bound assumes all of them fit into the same valleys at the same time.

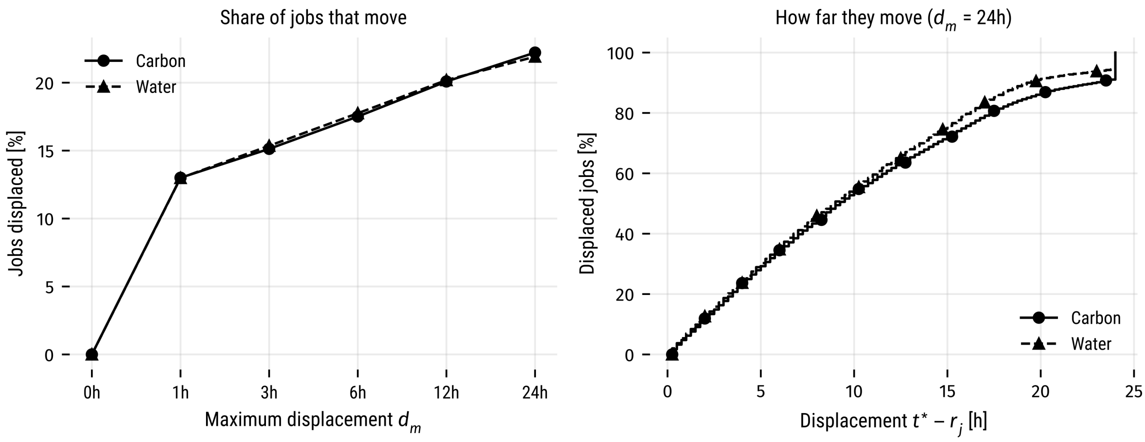

Figure 4.4: *Share of jobs the static analysis displaces as a function of d_m (left) and the distribution of the displacement $t^* - r_j$ among displaced jobs at a d_m of 24 hours (right).*

Shifting a job deterministically adds $t^* - r_j$ to its waiting time, so its bounded slowdown, $\max\big(1, (t^* - r_j + p_j)/\max(p_j, \tau)\big)$ with a bounding threshold of $\tau = 10$ s, rises with every displacement. Under the carbon objective at a d_m of 24 hours the mean bounded slowdown is 1.49 across all jobs and 3.22 among displaced jobs, with a 99th percentile of 15.4 (Figure 4.5). Saving and delay also move together across scenarios: per workload and window pair, the carbon saving correlates with the mean bounded slowdown at $r = 0.874$ (Figure 4.6). The windows that save the most are the windows where jobs wait the longest, so the trade-off is still there, but unlike under Greenfilling each unit of delay now buys a measured environmental return.

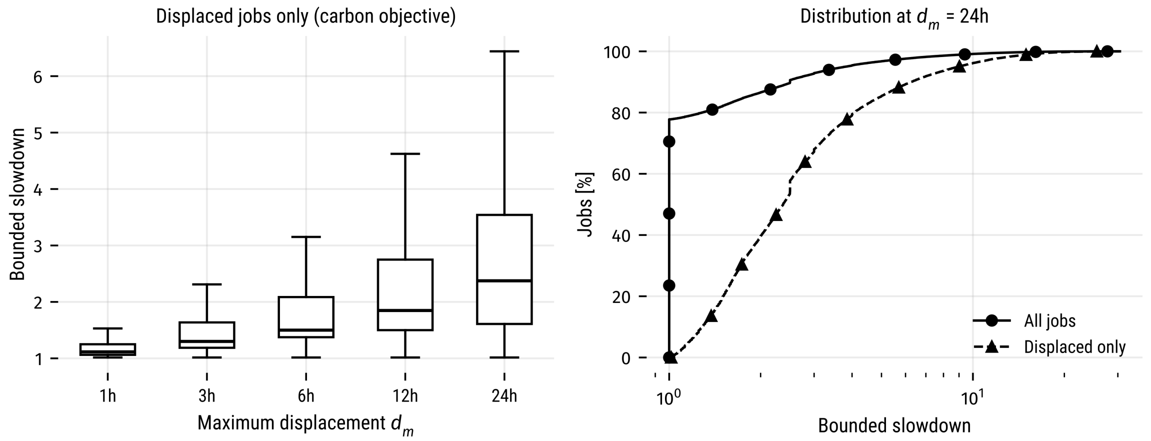

Figure 4.5: *Bounded slowdown implied by the displacements under the carbon objective. The left panel shows displaced jobs only, by maximum displacement. The right panel shows the cumulative distribution at a d_m of 24 hours for all jobs and for displaced jobs only.*

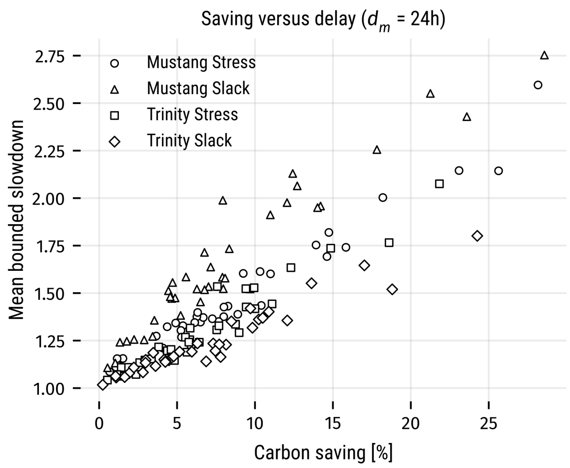

Figure 4.6: *Carbon saving against mean bounded slowdown at a d_m of 24 hours, one point per workload and window pair.*

4.2 Sufficiency-Aware Evaluation of HPC Systems

Chapter 5

Research Plan

5.1 Accounting for Environmental Cost in HPC Operations

5.2 Sufficiency-Aware Evaluation of HPC Systems

Chapter 6

Schedule and Expected Contributions

References

[Allen 2022] Michael Allen. "The huge carbon footprint of large-scale computing". *Physics World* 35.3 (Aug. 2022), pp. 46–50. issn: 0953-8585, 2058-7058. doi: 10.1088/2058-7058/35/03/32 (cit. on p. 1).

[Casanova *et al.* 2025] Henri Casanova, Arnaud Giersch, Arnaud Legrand, Martin Quinson, and Frédéric Suter. "Lowering entry barriers to developing custom simulators of distributed applications and platforms with SimGrid". *Parallel Computing* 123 (Mar. 2025), p. 103125. issn: 01678191. doi: 10.1016/j.parco.2025.103125 (cit. on p. 8).

[Czarnul *et al.* 2019] Pawel Czarnul, Jerzy Proficz, and Adam Krzywaniak. "Energy-Aware High-Performance Computing: Survey of State-of-the-Art Tools, Techniques, and Environments". *Scientific Programming* 2019 (Apr. 2019), pp. 1–19. issn: 1058-9244, 1875-919X. doi: 10.1155/2019/8348791 (cit. on p. 1).

[Dutot *et al.* 2017] Pierre-François Dutot, Michael Mercier, Millian Poquet, and Olivier Richard. "Batsim: a realistic language-independent resources and jobs management systems simulator". In: *Job Scheduling Strategies for Parallel Processing*. Ed. by Narayan Desai and Walfredo Cirne. Cham: Springer International Publishing, 2017, pp. 178–197. isbn: 978-3-319-61756-5 (cit. on p. 8).

[Feng and Cameron 2007] Wu-chun Feng and Kirk Cameron. "The Green500 List: Encouraging Sustainable Supercomputing". *Computer* 40.12 (Dec. 2007), pp. 50–55. issn: 0018-9162. doi: 10.1109/MC.2007.445 (cit. on p. 1).

[Gupta *et al.* 2021] Udit Gupta *et al.* "Chasing Carbon: The Elusive Environmental Footprint of Computing". In: *2021 IEEE International Symposium on High-Performance Computer Architecture (HPCA)*. Seoul, Korea (South): IEEE, Feb. 2021, pp. 854–867. isbn: 978-1-6654-2235-2. doi: 10.1109/HPCA51647.2021.00076 (cit. on p. 1).

[Hirth *et al.* 2018] Lion Hirth, Jonathan Mühlenpfordt, and Marisa Bulkeley. "The ENTSO-E Transparency Platform – A review of Europe's most ambitious electricity data platform". *Applied Energy* 225 (Sept. 2018), pp. 1054–1067. issn: 03062619. doi: 10.1016/j.apenergy.2018.04.048 (cit. on p. 8).

[Intergovernmental Panel On Climate Change 2015] Intergovernmental Panel On Climate Change. *Climate Change 2014: Mitigation of Climate Change: Working Group III Contribution to the IPCC Fifth Assessment Report.* 1st ed. Cambridge University Press, Jan. 2015. isbn: 978-1-107-05821-7 978-1-107-41541-6 978-1-107-65481-5. doi: 10.1017/CBO9781107415416 (cit. on p. 8).

[Kocot *et al.* 2023] Bartłomiej Kocot, Paweł Czarnul, and Jerzy Proficz. "Energy-Aware Scheduling for High-Performance Computing Systems: A Survey". *Energies* 16.2 (Jan. 2023), p. 890. issn: 1996-1073. doi: 10.3390/en16020890 (cit. on p. 1).

[Koomey *et al.* 2011] Jonathan Koomey, Stephen Berard, Marla Sanchez, and Henry Wong. "Implications of Historical Trends in the Electrical Efficiency of Computing". *IEEE Annals of the History of Computing* 33.3 (Mar. 2011), pp. 46–54. issn: 1058-6180, 1934-1547. doi: 10.1109/MAHC.2010.28 (cit. on p. 1).

[Li *et al.* 2023] Baolin Li *et al.* "Toward Sustainable HPC: Carbon Footprint Estimation and Environmental Implications of HPC Systems". In: *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis.* Denver CO USA: ACM, Nov. 2023, pp. 1–15. isbn: 979-8-4007-0109-2. doi: 10.1145/3581784.3607035 (cit. on p. 1).

[Macknick *et al.* 2012] J Macknick, R Newmark, G Heath, and K C Hallett. "Operational water consumption and withdrawal factors for electricity generating technologies: a review of existing literature". *Environmental Research Letters* 7.4 (Dec. 2012), p. 045802. issn: 1748-9326. doi: 10.1088/1748-9326/7/4/045802 (cit. on p. 8).

[Mu'alem and Feitelson 2001] A.W. Mu'alem and D.G. Feitelson. "Utilization, predictability, workloads, and user runtime estimates in scheduling the IBM SP2 with backfilling". *IEEE Transactions on Parallel and Distributed Systems* 12.6 (June 2001), pp. 529–543. issn: 10459219. doi: 10.1109/71.932708 (cit. on p. 7).

[Mytton 2021] David Mytton. "Data centre water consumption". *npj Clean Water* 4.1 (Feb. 2021), p. 11. issn: 2059-7037. doi: 10.1038/s41545-021-00101-w (cit. on p. 1).

[Pirson *et al.* 2023] Thibault Pirson *et al.* "The Environmental Footprint of IC Production: Review, Analysis, and Lessons From Historical Trends". *IEEE Transactions on Semiconductor Manufacturing* 36.1 (Feb. 2023), pp. 56–67. issn: 0894-6507, 1558-2345. doi: 10.1109/TSM.2022.3228311 (cit. on p. 1).

[Portegies Zwart 2020] Simon Portegies Zwart. "The ecological impact of high-performance computing in astrophysics". *Nature Astronomy* 4.9 (Sept. 2020), pp. 819–822. issn: 2397-3366. doi: 10.1038/s41550-020-1208-y (cit. on p. 1).

[Rosa *et al.* 2026] Lucas de Sousa Rosa *et al.* *The Environmental Impacts of High-Performance Computing: A Systematic Mapping Study.* 2026 (cit. on p. 1).

[Sandhu *et al.* 2025] Sofia Sandhu, Ayoub Zumeit, Zhenyu Tian, Vincenzo Vinciguerra, and Ravinder Dahiya. "Semiconductor manufacturing wastewater challenges and the potential solutions via printed electronics". *iScience* 28.10 (Oct. 2025), p. 113576. issn: 25890042. doi: 10.1016/j.isci.2025.113576 (cit. on p. 1).

[Saraiva *et al.* 2025] Gabriella Saraiva, Miguel Vasconcelos, Sarita Mazzini Bruschi, Danilo Carastan-Santos, and Daniel Cordeiro. *Estimating CO2 Emissions of Distributed Applications and Platforms with SimGrid/Batsim*. 2025. doi: 10.48550/ARXIV.2508.13693 (cit. on p. 8).

[Skovira *et al.* 1996] Joseph Skovira, Waiman Chan, Honbo Zhou, and David Lifka. "The EASY — LoadLeveler API project". In: *Job Scheduling Strategies for Parallel Processing*. Ed. by Gerhard Goos, Juris Hartmanis, J. Van Leeuwen, Dror G. Feitelson, and Larry Rudolph. Vol. 1162. Berlin, Heidelberg: Springer Berlin Heidelberg, 1996, pp. 41–47. isbn: 978-3-540-61864-5 978-3-540-70710-3. doi: 10.1007/BFb0022286 (cit. on pp. 2, 7).

[United Nations 2022] United Nations, ed. *Carbon Neutrality in the UNECE Region: Integrated Life-Cycle Assessment of Electricity Sources*. United Nations Publication. Geneva: United Nations, 2022. isbn: 978-92-1-001485-4 978-92-1-117298-0 (cit. on p. 8).

[York *et al.* 2022] Richard York, Lazarus Adua, and Brett Clark. "The rebound effect and the challenge of moving beyond fossil fuels: A review of empirical and theoretical research". *WIREs Climate Change* 13.4 (July 2022), e782. issn: 1757-7780, 1757-7799. doi: 10.1002/wcc.782 (cit. on p. 1).